

$$\frac{m_j \boldsymbol{r}_{j\alpha}^2}{\sum m_j \boldsymbol{r}_{j\alpha}^2} \longrightarrow \frac{m_1 \boldsymbol{r}_{11-2\parallel}^2 + m_2 \boldsymbol{r}_{21-2\parallel}^2}{\sum m_j \boldsymbol{r}_{j\alpha}^2} \quad (1)$$

$$\text{PED}_{1-2\alpha} = \frac{\frac{m_1 m_2}{m_1 + m_2} \left[\frac{(\boldsymbol{r}_{1\alpha} - \boldsymbol{r}_{2\alpha}) \boldsymbol{R}_{12}}{|\boldsymbol{R}_{12}|} \right]^2}{\sum m_j \boldsymbol{r}_{j\alpha}^2} \quad (2)$$

$$\boldsymbol{r}_1 \longrightarrow \boldsymbol{r}_{11-2} \longrightarrow \boldsymbol{r}_{11-2\parallel} \quad (3)$$

$$m_1 \boldsymbol{r}_{11-2\parallel}^2 + m_2 \boldsymbol{r}_{21-2\parallel}^2 = \frac{m_1 m_2}{m_1 + m_2} \left[\frac{(\boldsymbol{r}_1 - \boldsymbol{r}_2) \boldsymbol{R}_{12}}{|\boldsymbol{R}_{12}|} \right]^2 \quad (4)$$